The Dialogue Between Art and Technology: an Exploration of the Effectiveness of AISOMA and Its Application in Intangible Cultural Heritage Traditional Dance

Abstract

This paper aims to explore the effectiveness and potential of the Artificial Intelligence (AI) choreography tool AISOMA in the inheritance and innovation of Intangible Cultural Heritage (ICH) traditional dance. Faced with the challenges of globalization and modernization, the protection and inheritance of ICH traditional dance are of great significance. By adopting multi-dimensional research methods including literature analysis, case study, and assessment based on the APEC (Aesthetic, Practical, Emotional, and Cognitive) model, this study conducts an in-depth analysis of the technical characteristics of AISOMA and its application effects in ICH traditional dance. The research findings indicate that through deep learning mechanisms, style transfer technology, and real-time feedback mechanisms, AISOMA has effectively achieved the digital preservation of ICH traditional dance and promoted the innovative development of dance art. Moreover, AISOMA has not only enhanced the interactivity of dance education but also strengthened the emotional expression and cultural experience of dance works. Although AISOMA faces

challenges in terms of technological dependence, cultural adaptability, and ethical aspects, it has demonstrated tremendous potential in the protection, inheritance, and innovation of ICH traditional dance. The conclusion of this study points out that as an AI choreography tool, AISOMA provides a new direction for the future development of ICH traditional dance and is expected to play a more significant role in protecting and promoting human intangible cultural heritage.